20 March 1959
SUPERSEDING
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MILITARY SPECIFICATION

TRACER MARK 20 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, loading, assembly, and preparation for delivery of Tracer Mk. 20 Mod 0, and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

JAN-P-450 ——— Acid-Proof Black Paint, Types I and II.
MIL-A-148 ——— Aluminum Foil

STANDARDS

MILITARY

MIL-STD-105 ——— Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 ——— Marking for Shipment and Storage.
MIL-STD-414 ——— Sampling Procedures and Tables for Inspection by Variables.

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DRAWINGS

BUREAU OF ORDNANCE
L.D. 288823 ------ Tracer Mk. 20 Mod 0 List of Drawings, and all documents listed thereon.
Dwg. 985524 ------ Polyester Resin Sealant

PUBLICATIONS

BUREAU OF ORDNANCE
OP 400 ------ General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procurement Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 ------ Transportation, Interstate Commerce Commission, Explosives, and other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Revised 1950)").

3. REQUIREMENTS

3.1 General

3.1.1 Tracer Mk 20 Mod 0 shall be manufactured in accordance with List of Drawings 288823 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

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3.1.3 Preproduction sample. -- Unless otherwise specified in the contract or order, a preproduction sample of the tracer shall be manufactured using the methods and procedures proposed for the production lot. The preproduction sample will be tested as specified in section 4 herein and is for the purpose of determining that the production item meets the requirements of this specification. Provision of the sample shall be in accordance with 4.2.

3.2 Loading of Tracer

3.2.1 Loading of the tracer shall be in accordance with the applicable drawings. Loading may be carried out by scooping of the two mixes or by consolidation in the tracer-cup of the pre-pelleted compositions. However, pre-pelleting of the tracer-starter mixture is not recommended because of its sensitivity. Whichever method is employed, the weights of the tracer and tracer-starter pyrotechnic mixtures shall be as specified in 3.2.3.

3.2.2 The inside cavity of the tracer cup shall be coated with acid-proof black paint conforming to the requirements of Specification JAN-P-450, Type I or II, before loading. See applicable drawing.

3.2.3 Tracer Pyrotechnic -- Two increments, 1.15 grams each, of tracer pyrotechnic mixture -- R-324, shall be introduced into the tracer cup, previously coated with acid-proof black paint, and each increment pressed with a consolidating punch at a minimum pressure of 120,000 pounds per square inch for a minimum of three seconds. Then, one increment of 0.39 grams of tracer starter pyrotechnic mixture I-276 shall be introduced and pressed with the same consolidating punch at a minimum pressure of 120,000 pounds per square inch for a minimum of three seconds. The consolidating punch shall be in accordance with the applicable drawing. The height of the load shall be no less than 0.060 inch and no greater than 0.080 inch below the top of the cup.

3.2.3.1 All loose composition shall be removed from the surface of the load before sealing of the cup in 3.3.

3.2.3.2 If pre-formed pyrotechnic pellets are used instead of the powder increments specified above, the pellets shall be just hard enough to handle without excessive breakage. Pellets shall be made to weight and not to a fixed height. Reconsolidation of pellets shall be made to the pressures shown on the applicable drawings.

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3.3 Sealing of Tracer. -- After loading as in 3.2.3, an aluminum disc, 0.001 inch thick, shall be placed on the top of the load, and the lip of the tracer cup shall then be crimped 360° over the closing disc. The crimp seal and the closing disc shall then be coated with the sealant. See applicable drawing for details of the Polyester Resin Sealant.

3.4 Assembly of cup-tracer in projectile. -- The loaded and sealed tracer shall be assembled in projectiles in accordance with the applicable drawing for ballistic tests of 4.6.

3.4.1 The tracer-cavity of the projectile shall be coated with sufficient sealant so that, on insertion of the tracer by hand, the sealant is extruded somewhat. The tracer shall be held in place by a 360° staking of the rim of the projectile tracer-cavity. To insure the effectiveness of the seal on the tracer, additional sealant shall be applied after staking.

3.4.1.1 After insertion of the tracer in the projectile, the excess sealant is removed to prevent fouling of the staking punch during this operation. Proper regulation of the amount of sealant applied to the projectile tracer-cavity will insure effective filling of the spaces between the tracer and the projectile tracer-cavity and reduce to a minimum the cleaning operation to remove excess sealant. The sealant of tracer shall be permitted to cure 24 hours before firing projectiles.

3.5 Static Requirements. -- The tracer shall be tested as specified in 4.5.1.

3.5.1 The lower specification limit for static burning time shall be 6.0 seconds.

3.5.2 The lower specification limit for illuminating power shall be a minimum peak intensity of 150 foot-candles at two feet.

3.6 Ballistic Performance

3.6.1 Waterproofness Test.--The tracer shall be tested as specified in 4.6.1.

3.6.2 Tracer Functioning. -- The tracer shall have a minimum burning time of 1.10 seconds when tested as specified in 4.6.2.

3.7 Workmanship. -- Only the best trade practices shall be utilized in the manufacture of the Tracer Mk. 20 Mod 0. Care shall be exercised in the assembly of the component parts to assure proper seating and alignment especially where crimping is employed to maintain parts in their

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relative positions. There shall be no burrs or other imperfections of manufacture harmfully affecting the functioning or assembly of the product. All parts shall be free of chips, dirt, and other foreign materials. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. --- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. — As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot"; i.e., a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector,and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purposes.

4.2 Preproduction sample. — Before entering into quantity production, a preproduction sample of 1000 Tracers, Mk. 20 Mod 0, shall be delivered by the contractor for test to an agency designated by the Procuring Agency. These tracers shall be manufactured by the same procedures and processes and at the same location proposed by the contractor for the execution of the contract. When a preproduction sample is received which complies with all of the requirements of this specification and the appropriate drawings, the contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples shall become the property of the Procuring Agency and shall be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. — The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

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4.3 Production lot. -- For procurement purposes the first and second lots of tracers shall consist of 5,000 units plus those required for all test purposes. Subsequent lots shall consist of 50,000 units plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be provided at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.1.

4.4 Inspection. -- The general inspection procedures shall be in accordance with General instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400: Government inspection shall be in accordance with the provisions of General Procedures for Government Inspection, and contractor's inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Static tests

4.5.1 Static burning time and intensity Measurement. -- This test shall be performed using loaded, crimped-disc-sealed cup-tracers without the Laminac seal selected from each lot by the inspector. Laminac-sealed cup-tracers will not ignite under the conditions of this test. A small quantity of tracer-starter pyrotechnic, I-276, shall be placed on the crimped disc of the cup-tracer held in a vertical position. A piece of black-match shall be used to ignite the system. The instrumentation for determination of this illumination shall be protected from molten slag, (which can travel up to four feet) either by distance or by an air curtain or both. Computation of the lower illumination intensity requirement, (for distances other than two feet) may be by the following formula:

$$\text{Minimum peak illumination required at a given distance} = \frac{86400 \text{ ft.-candles}}{(\text{Distance in inches})^2}$$

4.5.1.1 Sampling and Acceptance Criteria, Static Burning time and Intensity. — Sample size and acceptance criteria for the static burning time and intensity test shall be determined in accordance with MIL-STD-414, AQL of 1%. Lower specification limits shall be as indicated in 3.5.1. and 3.5.2. Failure of a sample to comply with these requirements shall result in the rejection of the lot represented.

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4.6 Ballistic Tests. -- Samples for gun firing tests shall be selected from each production lot by the Government Inspector and forwarded to the Station or Proving Ground, designated by the Procuring agency for ballistic tests. The tests shall be conducted in a weapon designated by the Procuring Agency. Suitable aids, such as binoculars, should be used to insure that the last instant of trace is obtained.

4.6.1 Waterproofness Test. -- A sample of tracers with the specified seal shall be selected by the inspector from each lot. They shall be immersed in water to a depth of not less than 2 inches nor more than 3 inches, after bringing both the tracers and the water to room temperature, and allowed to stand for a period of not less than 48 hours. Immediately after immersion, the tracers shall be assembled into projectiles as specified in 3.4 and test fired in a weapon in single action, allowing time specified in 3.4.1.1 for sealant to dry before firing. The burning time of each tracer round shall be measured and recorded. Duds or units characterized by burning times less than 1.1 seconds shall be noted and reported. Only duds are to be considered as basis for rejection of lot under acceptance criteria of 4.6.1.1.

4.6.1.1 Sampling and Acceptance Criteria, Waterproofness Test. -- Compliance with the waterproofness test for duds shall be determined in accordance with MIL-STD-105-A, Code Letter "H", double sampling plan, AQL of 4%.

Sample	Sample Size	Cumulative Sample Size	Accept No.	Reject No.
1st[1]	25	25	2 (Duds)	5 (Duds)
2nd	50	75	4 (Duds)	5 (Duds)

[1]Acceptance may be made on first sample if twenty-three (23) units are satisfactory regardless of whether the two (2) unsatisfactory units are duds or blown tracers.

4.6.2 Tracer Functioning Tests. -- Tracers shall be selected from each production lot, assembled into projectiles as specified in 3.4 and subjected to the specified gun firing tests.

4.6.2.1 Tracer Proof Round Test. -- Five (5) rounds shall be fired at room temperature in single action with a proof charge of propellant. Tracer functioning times (total time of visible trace) shall be measured and recorded. The number of duds and/or burning times recorded shall be forwarded to the Procuring Agency for information purposes.

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4.6.2.2 Tracer Service Round test. — Tracers randomly selected from each lot shall be assembled in service rounds and fired, single action, at room temperature. Sample size and acceptance criteria for duds shall be determined in accordance with MIL-STD-105A, double sampling plan, code letter "I" AQL 2.5% defective. Evaluation of compliance with a lower burn time requirement of 1.1 seconds shall be determined in accordance with MIL-STD-414 code letter "K" AQL 4.0% defective. The first sample (35 units) of the above double sampling plan may be used to satisfy the variable sample size requirements of the MIL-STD-414. If a total of one dud or blowout occurs in the first sample (35 units) burning time may be evaluated on the basis of 34 observations.

4.6.3 The ballistic test results shall be reported to the interested Government agency on an appropriate form, giving all necessary data, including mean burning time, standard deviation of the burning time, lot quantity index, number of short burning times, and number of duds.

4.6.4 Invalid Tests. — Should a sample fail to comply with the specified ballistic test requirements, a second test shall be authorized only if the failures can be ascribed to deficiencies in the testing conditions. In such event the first test shall be discounted and the results of the second test shall be used to determine lot disposition.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Tracers shall be packaged in accordance with applicable drawings.

5.2 Packing. — The packages of tracers shall be packed in accordance with applicable drawings.

5.3 Marking. — In addition to any special marking required by the contract or order, each container shall be marked in accordance with applicable drawings.

5.4 Data card. — Tracer data cards, 5" by 8", made of commercial tag board, approximately 200 pounds ream weight, and printed as shown on the form below, shall be properly filled out for each lot of accepted tracers. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Procuring Activity.

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TRACER DATA CARD

Tracer_____ Tracer Lot No._____

Quantity in Lot _____

Drg. No._____ Spec. No._____

Metal Parts Manufactured by _____Lot No._____

Loaded and Assembled by_____Date_____

Contract No._____ Order No._____

Packed_____ in a Box.

Remarks:

Certified to by _____ Inspector

(Print here name of Inspector, District, Arsenal, or Loading Plant, and Sub-district where applicable.)

5.4.1 Tracer Lot No. -- An identification shall be given to each production lot as follows:

Manufacturer's Initials	- Lot No. (beginning with 1 for each calendar year) -	Date of Loading (Month & year)

6. NOTES

6.1 Intended use. -- Tracer Mk. 20 Mod 0 is designed to show a bright trace of reddish light such that the trajectory of the projectile in which it is mounted may be clearly observed.

6.2 Ordering data. -- Procurement documents should specify the title, symbol, and date of this specification.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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